The Bias–Variance Trade-off in Ridge Regression

A template demonstration: mathematics, figures, code and cross-references

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October 10, 2026

ABSTRACT We use ridge regression to show what the template looks like. Under the Gaussian linear model we derive the bias and variance of the ridge estimator, explain how a little shrinkage trades a small bias for a large drop in variance, and check this with a simulation.

Keywords ridge regression; bias–variance decomposition; shrinkage

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1 Introduction

When the columns of the design matrix X are strongly correlated, the variance of the least-squares estimator $\hat{\beta} = (X^\top X)^{-1} X^\top y$ explodes. Hoerl and Kennard (1970) proposed adding a positive constant to the diagonal of $X^\top X$, *giving up unbiasedness* in exchange for a far more stable estimate; the idea became a cornerstone of shrinkage estimation (Hastie et al. 2009).[1] Section 2 sets up the model and Section 3 gives a numerical example.

2 Model and estimator

2.1 Setting

Assumption 1 (Gaussian linear model)**.** We observe $y = X\beta + \varepsilon$, where $X \in \mathbb{R}^{n \times p}$ is fixed and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$.

Definition 1 (Ridge estimator)**.** For $\lambda > 0$ the ridge estimator is

$$\hat{\beta}_\lambda = \arg\min_{b \in \mathbb{R}^p} \left\{ \|y - Xb\|_2^2 + \lambda \|b\|_2^2 \right\} = (X^\top X + \lambda I_p)^{-1} X^\top y. \tag{1}$$

2.2 Bias and variance

Write $X = UDV^\top$ for the singular value decomposition, with $d_1 \geq \cdots \geq d_p > 0$.

[1] Every number in this note can be reproduced with the code in Section 3.

Proposition 1. *Under Assumption 1, the bias and covariance of the ridge estimator are*

$$\operatorname{Bias}(\hat{\beta}_\lambda) = -\lambda\,(X^\top X + \lambda I_p)^{-1}\beta, \tag{2}$$

$$\operatorname{Var}(\hat{\beta}_\lambda) = \sigma^2 \sum_{j=1}^{p} \frac{d_j^2}{(d_j^2+\lambda)^2}\, v_j v_j^\top. \tag{3}$$

Hence $\operatorname{tr}\operatorname{Var}(\hat{\beta}_\lambda)$ *is strictly decreasing in* λ*, while* $\|\operatorname{Bias}(\hat{\beta}_\lambda)\|_2^2$ *is strictly increasing whenever* $\beta \neq 0$*.*

Proof. By Equation (1) and $\mathbb{E}y = X\beta$, $\mathbb{E}\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top X \beta$; subtracting β gives Equation (2). Since $\operatorname{Var}(y) = \sigma^2 I_n$, substituting $(X^\top X + \lambda I)^{-1} X^\top = V\,\operatorname{diag}(d_j/(d_j^2+\lambda))U^\top$ gives Equation (3). □

Remark 1. As $\lambda \to 0$, $\hat{\beta}_\lambda$ tends to the least-squares estimator; as $\lambda \to \infty$, $\hat{\beta}_\lambda \to 0$. The risk $\operatorname{MSE}(\lambda) = \|\operatorname{Bias}\|_2^2 + \operatorname{tr}\operatorname{Var}$ is minimised in between.

3 A numerical example

Take $n = 120$, $p = 6$, rows of X drawn from $\mathcal{N}(0, \Sigma)$ with $\Sigma_{ij} = 0.6^{|i-j|}$, $\beta = (2, -1.5, 1, 0, 0, 0.5)^\top$ and $\sigma = 2$. Figure 1 shows how the coefficients shrink as λ grows, and Table 1 gives the exact risk decomposition from Equations (2) and (3).

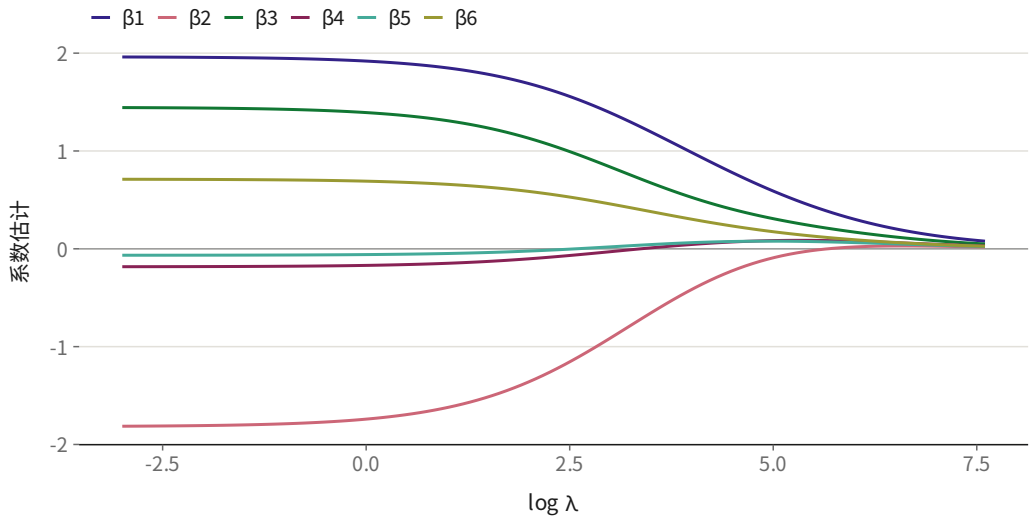


Figure 1. Ridge coefficient paths. As λ grows, every coefficient is pulled continuously towards zero.

Table 1 makes three points:

1. the variance falls and the bias rises monotonically, as Proposition 1 predicts;
2. around $\lambda = 2$ the MSE is below that of least squares ($\lambda = 0$);
3. heavy shrinkage ($\lambda = 100$) lets the bias dominate the risk.

Computing the estimator takes one line of linear algebra:

Table 1. Bias, variance and mean squared error for several values of λ

λ	$\lVert\text{Bias}\rVert_2^2$	$\operatorname{tr}\text{Var}$	MSE
0	0.000	0.449	0.449
1	0.008	0.424	0.432
2	0.029	0.402	**0.430**
5	0.148	0.345	0.493
20	1.110	0.195	1.305
100	4.030	0.048	4.078

```
ridge <- function(X, y, lambda) {
  p <- ncol(X)
  solve(crossprod(X) + lambda * diag(p), crossprod(X, y))  # normal equations
}
```

4 Conclusion

Ridge regression illustrates a recurring theme in statistics: *unbiased is not the same as good*. In practice λ is chosen by cross-validation.

References

Hastie, T., R. Tibshirani, and J. Friedman (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd ed. New York: Springer. DOI: 10.1007/978-0-387-84858-7.

Hoerl, A. E. and R. W. Kennard (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics* 12.1, pp. 55–67. DOI: 10.1080/00401706.1970.10488634.

A An extra derivation

For an orthogonal design $X^\top X = I_p$ the ridge estimator is a rescaled least-squares estimator:

$$\hat\beta_\lambda = \frac{1}{1+\lambda}\,\hat\beta^{\text{OLS}}, \qquad \text{MSE}(\lambda) = \frac{p\,\sigma^2 + \lambda^2 \lVert\beta\rVert_2^2}{(1+\lambda)^2},$$

which is minimised at $\lambda^\star = p\sigma^2 / \lVert\beta\rVert_2^2$.